Remarks

Many of the properties of the real system hold in the complex number system, but there are some remarkable differences as well. For example, we cannot compare two complex numbers $z_1 = x_1 + iy_1$, $y_1 \neq 0$, and $z_2 = x_2 + iy_2$, $y_2 \neq 0$, by means of inequalities. In other words, statements such as $z_1 < z_2$ and $z_2 \geq z_1$ have no meaning except in the case when the two numbers z_1 and z_2 are real. We can, however, compare the absolute values of two complex numbers. Thus, if $z_1 = 3 + 4i$ and $z_2 = 5 - i$, then $|z_1| = 5$ and $|z_2| = \sqrt{26}$, and consequently $|z_1| < |z_2|$. This last inequality means that the point $(3, 4)$ is closer to the origin than is the point $(5, -1)$.

17.1 Exercises

Answers to selected odd-numbered problems begin on page ANS-38.

In Problems 1–26, write the given number in the form $a + ib$.

1. $2i^3 - 3i^2 + 5i$

2. $3i^5 - i^4 + 7i^3 - 10i^2 - 9$

3. i^8

4. i^{11}

5. $(5 - 9i) + (2 - 4i)$

6. $3(4 - i) - 3(5 + 2i)$

7. $i(5 + 7i)$

8. $i(4 - i) + 4i(1 + 2i)$

9. $(2 - 3i)(4 + i)$

10. $(\frac{1}{2} - \frac{1}{4}i)(\frac{2}{3} + \frac{5}{3}i)$

11. $(2 + 3i)^2$

12. $(1 - i)^3$

13. $\dfrac{2}{i}$

14. $\dfrac{i}{1 + i}$

15. $\dfrac{2 - 4i}{3 + 5i}$

16. $\dfrac{10 - 5i}{6 + 2i}$

17. $\dfrac{(3 - i)(2 + 3i)}{1 + i}$

18. $\dfrac{(1 + i)(1 - 2i)}{(2 + i)(4 - 3i)}$

19. $\dfrac{(5 - 4i) - (3 + 7i)}{(4 + 2i) + (2 - 3i)}$

20. $\dfrac{(4 + 5i) + 2i^3}{(2 + i)^2}$

21. $i(1 - i)(2 - i)(2 + 6i)$

22. $(1 + i)^2(1 - i)^3$

23. $(3 + 6i) + (4 - i)(3 + 5i) + \dfrac{1}{2 - i}$

24. $(2 + 3i)\left(\dfrac{2 - i}{1 + 2i}\right)^2$

25. $\left(\dfrac{i}{3 - i}\right)\left(\dfrac{1}{2 + 3i}\right)$

26. $\dfrac{1}{(1 + i)(1 - 2i)(1 + 3i)}$

In Problems 27–32, let $z = x + iy$. Find the indicated expression.

27. $\text{Re}(1/z)$

28. $\text{Re}(z^2)$

29. $\text{Im}(2z + 4\bar{z} - 4i)$

30. $\text{Im}(\bar{z}^2 + z^2)$

31. $|z - 1 - 3i|$

32. $|z + 5\bar{z}|$

In Problems 33–38, use Definition 17.1.2 to find a complex number z satisfying the given equation.

33. $2z = i(2 + 9i)$

34. $z - 2\bar{z} + 7 - 6i = 0$

35. $z^2 = i$

36. $\bar{z}^2 = 4z$

37. $z + 2\bar{z} = \dfrac{2 - i}{1 + 3i}$

38. $\dfrac{z}{1 + \bar{z}} = 3 + 4i$

In Problems 39 and 40, determine which complex number is closer to the origin.

39. $10 + 8i$, $11 - 6i$

40. $\frac{1}{2} - \frac{1}{4}i$, $\frac{2}{3} + \frac{1}{6}i$

41. Prove that $|z_1 - z_2|$ is the distance between the points z_1 and z_2 in the complex plane.

42. Show for all complex numbers z on the circle $x^2 + y^2 = 4$ that $|z + 6 + 8i| \leq 12$.

For Discussion

43. For n a nonnegative integer, i^n can be one of four values: i, -1, $-i$, and 1. In each of the following four cases express the integer exponent n in terms of the symbol k, where $k = 0, 1, 2, \ldots$.
(a) $i^n = i$ **(b)** $i^n = -1$ **(c)** $i^n = -i$ **(d)** $i^n = 1$

44. (a) Without doing any significant work such as multiplying out or using the binomial theorem, think of an easy way of evaluating $(1 + i)^8$.
(b) Use your method in part (a) to evaluate $(1 + i)^{64}$.

17.2 Powers and Roots

Introduction Recall from calculus that a point (x, y) in rectangular coordinates can also be expressed in terms of polar coordinates (r, θ). We shall see in this section that the ability to express a complex number z in terms of r and θ greatly facilitates finding powers and roots of z.

Polar Form Rectangular coordinates (x, y) and polar coordinates (r, θ) are related by the equations $x = r\cos\theta$ and $y = r\sin\theta$ (see Section 14.1). Thus a nonzero complex number $z = x + iy$ can be written as $z = (r\cos\theta) + i(r\sin\theta)$ or

$$z = r(\cos\theta + i\sin\theta). \tag{1}$$

We say that (1) is the **polar form** of the complex number z. We see from **FIGURE 17.2.1** that the polar coordinate r can be interpreted as the distance from the origin to the point (x, y). In other words, we adopt the convention that r is never negative so that we can take r to be the modulus of z; that is, $r = |z|$. The angle θ of inclination of the vector z measured in radians from the positive real axis is positive when measured counterclockwise and negative when measured clockwise. The angle θ is called an **argument** of z and is written $\theta = \arg z$. From Figure 17.2.1 we see that an argument of a complex number must satisfy the equation $\tan\theta = y/x$. The solutions of this equation are not unique, since if θ_0 is an argument of z, then necessarily the angles $\theta_0 \pm 2\pi$, $\theta_0 \pm 4\pi, \ldots$, are also arguments. The argument of a complex number in the interval $-\pi < \theta \le \pi$ is called the **principal argument** of z and is denoted by Arg z. For example, $\operatorname{Arg}(i) = \pi/2$.

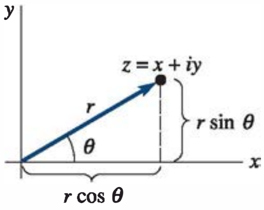

FIGURE 17.2.1 Polar coordinates

EXAMPLE 1 A Complex Number in Polar Form

Express $1 - \sqrt{3}i$ in polar form.

SOLUTION With $x = 1$ and $y = -\sqrt{3}$, we obtain $r = |z| = \sqrt{(1)^2 + (-\sqrt{3})^2} = 2$. Now since the point $(1, -\sqrt{3})$ lies in the fourth quadrant, we can take the solution of $\tan\theta = -\sqrt{3}/1 = -\sqrt{3}$ to be $\theta = \arg z = 5\pi/3$. It follows from (1) that a polar form of the number is

$$z = 2\left(\cos\frac{5\pi}{3} + i\sin\frac{5\pi}{3}\right).$$

As we see in **FIGURE 17.2.2**, the argument of $1 - \sqrt{3}i$ that lies in the interval $(-\pi, \pi]$, the principal argument of z, is $\operatorname{Arg} z = -\pi/3$. Thus, an alternative polar form of the complex number is

$$z = 2\left[\cos\left(-\frac{\pi}{3}\right) + i\sin\left(-\frac{\pi}{3}\right)\right].$$ ≡

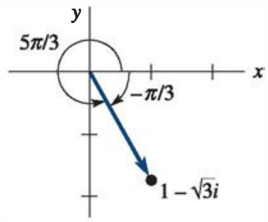

FIGURE 17.2.2 Two arguments of $z = 1 - \sqrt{3}i$ in Example 1

Multiplication and Division

The polar form of a complex number is especially convenient to use when multiplying or dividing two complex numbers. Suppose

$$z_1 = r_1(\cos\theta_1 + i\sin\theta_1) \quad \text{and} \quad z_2 = r_2(\cos\theta_2 + i\sin\theta_2),$$

where θ_1 and θ_2 are any arguments of z_1 and z_2, respectively. Then

$$z_1z_2 = r_1r_2[(\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) + i(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2)] \tag{2}$$

and for $z_2 \neq 0$,

$$\frac{z_1}{z_2} = \frac{r_1}{r_2}[(\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2) + i(\sin\theta_1\cos\theta_2 - \cos\theta_1\sin\theta_2)]. \tag{3}$$

From the addition formulas from trigonometry, (2) and (3) can be rewritten, in turn, as

$$z_1z_2 = r_1r_2[\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)] \tag{4}$$

and

$$\frac{z_1}{z_2} = \frac{r_1}{r_2}[\cos(\theta_1 - \theta_2) + i\sin(\theta_1 - \theta_2)]. \tag{5}$$

Inspection of (4) and (5) shows that

$$|z_1z_2| = |z_1||z_2|, \quad \left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}, \tag{6}$$

and

$$\arg(z_1z_2) = \arg z_1 + \arg z_2, \quad \arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2. \tag{7}$$

EXAMPLE 2 Argument of a Product and of a Quotient

We have seen that $\operatorname{Arg} z_1 = \pi/2$ for $z_1 = i$. In Example 1 we saw that $\operatorname{Arg} z_2 = -\pi/3$ for $z_2 = 1 - \sqrt{3}i$. Thus, for

$$z_1z_2 = i(1 - \sqrt{3}i) = \sqrt{3} + i \quad \text{and} \quad \frac{z_1}{z_2} = \frac{i}{1 - \sqrt{3}i} = -\frac{\sqrt{3}}{4} + \frac{1}{4}i$$

it follows from (7) that

$$\arg(z_1z_2) = \frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{6} \quad \text{and} \quad \arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{2} - \left(-\frac{\pi}{3}\right) = \frac{5\pi}{6}.$$

≡

In Example 2 we used the principal arguments of z_1 and z_2 and obtained $\arg(z_1z_2) = \operatorname{Arg}(z_1z_2)$ and $\arg(z_1/z_2) = \operatorname{Arg}(z_1/z_2)$. It should be observed, however, that this was a coincidence. Although (7) is true for any arguments of z_1 and z_2, *it is not true*, in general, that $\operatorname{Arg}(z_1z_2) = \operatorname{Arg} z_1 + \operatorname{Arg} z_2$ and $\operatorname{Arg}(z_1/z_2) = \operatorname{Arg} z_1 - \operatorname{Arg} z_2$. See Problem 39 in Exercises 17.2.

Powers of z We can find integer powers of the complex number z from the results in (4) and (5). For example, if $z = r(\cos\theta + i\sin\theta)$, then with $z_1 = z$ and $z_2 = z$, (4) gives

$$z^2 = r^2[\cos(\theta+\theta) + i\sin(\theta+\theta)] = r^2(\cos 2\theta + i\sin 2\theta).$$

Since $z^3 = z^2z$, it follows that

$$z^3 = r^3(\cos 3\theta + i\sin 3\theta).$$

Moreover, since $\arg(1) = 0$, it follows from (5) that

$$\frac{1}{z^2} = z^{-2} = r^{-2}[\cos(-2\theta) + i\sin(-2\theta)].$$

Continuing in this manner, we obtain a formula for the nth power of z for any integer n:

$$z^n = r^n(\cos n\theta + i\sin n\theta). \tag{8}$$

EXAMPLE 3 Power of a Complex Number

Compute z^3 for $z = 1 - \sqrt{3}i$.

SOLUTION In Example 1 we saw that

$$z = 2\left[\cos\left(-\frac{\pi}{3}\right) + i\sin\left(-\frac{\pi}{3}\right)\right].$$

Hence from (8) with $r = 2$, $\theta = -\pi/3$, and $n = 3$, we get

$$\begin{aligned}(1-\sqrt{3}i)^3 &= 2^3\left[\cos\left(3\left(-\frac{\pi}{3}\right)\right) + i\sin\left(3\left(-\frac{\pi}{3}\right)\right)\right]\\ &= 8[\cos(-\pi) + i\sin(-\pi)] = -8.\end{aligned}$$

≡

DeMoivre's Formula When $z = \cos\theta + i\sin\theta$, we have $|z| = r = 1$ and so (8) yields

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta. \tag{9}$$

This last result is known as **DeMoivre's formula** and is useful in deriving certain trigonometric identities. See Problems 37 and 38 in Exercises 17.2.

Roots A number w is said to be an ***n*th root** of a nonzero complex number z if $w^n = z$. If we let $w = \rho(\cos\phi + i\sin\phi)$ and $z = r(\cos\theta + i\sin\theta)$ be the polar forms of w and z, then in view of (8), $w^n = z$ becomes

$$\rho^n(\cos n\phi + i\sin n\phi) = r(\cos\theta + i\sin\theta).$$

From this we conclude that $\rho^n = r$ or $\rho = r^{1/n}$ and

$$\cos n\phi + i\sin n\phi = \cos\theta + i\sin\theta.$$

By equating the real and imaginary parts, we get from this equation

$$\cos n\phi = \cos\theta \quad \text{and} \quad \sin n\phi = \sin\theta.$$

These equalities imply that $n\phi = \theta + 2k\pi$, where k is an integer. Thus,

$$\phi = \frac{\theta + 2k\pi}{n}.$$

As k takes on the successive integer values $k = 0, 1, 2, \ldots, n - 1$, we obtain n *distinct* roots with the same modulus but different arguments. But for $k \geq n$ we obtain the same roots because the sine and cosine are 2π-periodic. To see this, suppose $k = n + m$, where $m = 0, 1, 2, \ldots$. Then

$$\phi = \frac{\theta + 2(n + m)\pi}{n} = \frac{\theta + 2m\pi}{n} + 2\pi$$

and so $$\sin\phi = \sin\left(\frac{\theta + 2m\pi}{n}\right), \qquad \cos\phi = \cos\left(\frac{\theta + 2m\pi}{n}\right).$$

We summarize this result. The n nth roots of a nonzero complex number $z = r(\cos\theta + i\sin\theta)$ are given by

$$w_k = r^{1/n}\left[\cos\left(\frac{\theta + 2k\pi}{n}\right) + i\sin\left(\frac{\theta + 2k\pi}{n}\right)\right], \tag{10}$$

where $k = 0, 1, 2, \ldots, n - 1$.

EXAMPLE 4 Roots of a Complex Number

Find the three cube roots of $z = i$.

SOLUTION With $r = 1$, $\theta = \arg z = \pi/2$, the polar form of the given number is $z = \cos(\pi/2) + i\sin(\pi/2)$. From (10) with $n = 3$ we obtain

$$w_k = (1)^{1/3}\left[\cos\left(\frac{\pi/2 + 2k\pi}{3}\right) + i\sin\left(\frac{\pi/2 + 2k\pi}{3}\right)\right], \quad k = 0, 1, 2.$$

Hence, the three roots are

$$k = 0, \quad w_0 = \cos\frac{\pi}{6} + i\sin\frac{\pi}{6} = \frac{\sqrt{3}}{2} + \frac{1}{2}i$$

$$k = 1, \quad w_1 = \cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6} = -\frac{\sqrt{3}}{2} + \frac{1}{2}i$$

$$k = 2, \quad w_2 = \cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2} = -i.$$

≡

The root w of a complex number z obtained by using the principal argument of z with $k = 0$ is sometimes called the **principal nth root** of z. In Example 4, since $\mathrm{Arg}(i) = \pi/2$, $w_0 = \sqrt{3}/2 + (1/2)i$ is the principal third root of i.

Since the roots given by (8) have the same modulus, the n roots of a nonzero complex number z lie on a circle of radius $r^{1/n}$ centered at the origin in the complex plane. Moreover, since the difference between the arguments of any two successive roots is $2\pi/n$, the nth roots of z are equally spaced on this circle. **FIGURE 17.2.3** shows the three cube roots of i equally spaced on a unit circle; the angle between roots (vectors) w_k and w_{k+1} is $2\pi/3$.

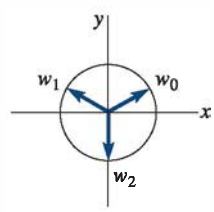

FIGURE 17.2.3 Three cube roots of i

As the next example will show, the roots of a complex number do not have to be "nice" numbers as in Example 3.

EXAMPLE 5 Roots of a Complex Number

Find the four fourth roots of $z = 1 + i$.

SOLUTION In this case, $r = \sqrt{2}$ and $\theta = \arg z = \pi/4$. From (10) with $n = 4$, we obtain

$$w_k = (\sqrt{2})^{1/4}\left[\cos\left(\frac{\pi/4 + 2k\pi}{4}\right) + i\sin\left(\frac{\pi/4 + 2k\pi}{4}\right)\right], \quad k = 0, 1, 2, 3.$$

The roots, rounded to four decimal places, are

$$k = 0, \quad w_0 = (\sqrt{2})^{1/4}\left[\cos\frac{\pi}{16} + i\sin\frac{\pi}{16}\right] = 1.0696 + 0.2127i$$

$$k = 1, \quad w_1 = (\sqrt{2})^{1/4}\left[\cos\frac{9\pi}{16} + i\sin\frac{9\pi}{16}\right] = -0.2127 + 1.0696i$$

$$k = 2, \quad w_2 = (\sqrt{2})^{1/4}\left[\cos\frac{17\pi}{16} + i\sin\frac{17\pi}{16}\right] = -1.0696 - 0.2127i$$

$$k = 3, \quad w_3 = (\sqrt{2})^{1/4}\left[\cos\frac{25\pi}{16} + i\sin\frac{25\pi}{16}\right] = 0.2127 - 1.0696i.$$

≡

17.2 Exercises

Answers to selected odd-numbered problems begin on page ANS-38.

In Problems 1–10, write the given complex number in polar form.

1. 2
2. -10
3. $-3i$
4. $6i$
5. $1 + i$
6. $5 - 5i$
7. $-\sqrt{3} + i$
8. $-2 - 2\sqrt{3}i$
9. $\dfrac{3}{-1 + i}$
10. $\dfrac{12}{\sqrt{3} + i}$

In Problems 11–14, write the number given in polar form in the form $a + ib$.

11. $z = 5\left(\cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6}\right)$

12. $z = 8\sqrt{2}\left(\cos\frac{11\pi}{4} + i\sin\frac{11\pi}{4}\right)$

13. $z = 6\left(\cos\frac{\pi}{8} + i\sin\frac{\pi}{8}\right)$

14. $z = 10\left(\cos\frac{\pi}{5} + i\sin\frac{\pi}{5}\right)$

In Problems 15 and 16, find z_1z_2 and z_1/z_2. Write the number in the form $a + ib$.

15. $z_1 = 2\left(\cos\frac{\pi}{8} + i\sin\frac{\pi}{8}\right), z_2 = 4\left(\cos\frac{3\pi}{8} + i\sin\frac{3\pi}{8}\right)$

16. $z_1 = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right),$

$z_2 = \sqrt{3}\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right)$

In Problems 17–20, write each complex number in polar form. Then use either (4) or (5) to obtain a polar form of the given number. Write the polar form in the form $a + ib$.

17. $(3 - 3i)(5 + 5\sqrt{3}i)$
18. $(4 + 4i)(-1 + i)$
19. $\dfrac{-i}{2 - 2i}$
20. $\dfrac{\sqrt{2} + \sqrt{6}i}{-1 + \sqrt{3}i}$

In Problems 21–26, use (8) to compute the indicated power.

21. $(1 + \sqrt{3}i)^9$
22. $(2 - 2i)^5$
23. $(\frac{1}{2} + \frac{1}{2}i)^{10}$
24. $(-\sqrt{2} + \sqrt{6}i)^4$

25. $\left(\cos\frac{\pi}{8} + i\sin\frac{\pi}{8}\right)^{12}$

26. $\left[\sqrt{3}\left(\cos\frac{2\pi}{9} + i\sin\frac{2\pi}{9}\right)\right]^6$

In Problems 27–32, use (10) to compute all roots. Sketch these roots on an appropriate circle centered at the origin.

27. $(8)^{1/3}$
28. $(1)^{1/8}$
29. $(i)^{1/2}$
30. $(-1 + i)^{1/3}$
31. $(-1 + \sqrt{3}i)^{1/2}$
32. $(-1 - \sqrt{3}i)^{1/4}$

In Problems 33 and 34, find all solutions of the given equation.

33. $z^4 + 1 = 0$
34. $z^8 - 2z^4 + 1 = 0$

In Problems 35 and 36, express the given complex number first in polar form and then in the form $a + ib$.

35. $\left(\cos\frac{\pi}{9} + i\sin\frac{\pi}{9}\right)^{12}\left[2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)\right]^5$

36. $\dfrac{\left[8\left(\cos\frac{3\pi}{8} + i\sin\frac{3\pi}{8}\right)\right]^3}{\left[2\left(\cos\frac{\pi}{16} + i\sin\frac{\pi}{16}\right)\right]^{10}}$

37. Use the result $(\cos\theta + i\sin\theta)^2 = \cos 2\theta + i\sin 2\theta$ to find trigonometric identities for $\cos 2\theta$ and $\sin 2\theta$.

38. Use the result $(\cos\theta + i\sin\theta)^3 = \cos 3\theta + i\sin 3\theta$ to find trigonometric identities for $\cos 3\theta$ and $\sin 3\theta$.